Dual-Task Cascaded Network for Spatial–Temporal–Spectral Remote Sensing Image Fusion

Xiangchao Meng, *Senior Member, IEEE*, Xu Chen, Mengjing Zhang, Feng Shao, *Member, IEEE*, Gang Yang, Weiwei Sun, *Senior Member, IEEE*, and Liang Chen

***Abstract*— Spatial–temporal–spectral fusion (STSF) is dedicated to integrating the complementary advantages of multisource images to obtain fused images with all high spatial, high temporal, and high spectral resolutions, which is promising but more challenging. On the one hand, traditional studies deployed on MODIS and Landsat data cannot be transferred to most spaceborne hyperspectral (HS) data with lower temporal resolution; on the other hand, the rigid time relation modeling in most existing studies exhibits weakness orienting to nonlinear land-cover changes. In this article, we propose a dual-task cascaded network for STSF (DC-STSF), with collaborative modeling on spatial–spectral joint enhancement (SSJE) and temporal variation estimation (TVE) in a unified framework. The SSJE task was designed with an iterative alternating projection, meticulously crafted to address the scale variance among observations. Additionally, the spatial enhancement unit (SEU) and error correction unit (ECU) were coupled modeling to enhance the spatial and spectral fidelities. The TVE on the spectral fine-tuning network (SFTN) was developed, to further enhance the temporal and spectral fidelities. Extensive experiments were implemented on Ziyuan (ZY)-1 02D HS data and Sentinel-2 multispectral (MS) data. Both qualitative and quantitative results demonstrated the competitive performance of the proposed method.**

***Index Terms*— Hyperspectral (HS) image, multispectral (MS) image, remote sensing, spatial–temporal–spectral fusion (STSF), temporal change.**

I. Introduction

THE remote sensing images carrying all high spatial, temporal, and spectral resolutions have contributed greatly to varieties of applications, such as disaster warning [1], object detection [2], land-cover classification [3], and so on [4], [5], [6]. Unfortunately, due to hardware limitations and other factors, there is a fundamental tradeoff among the spatial, temporal, and spectral resolutions. That is as the ancient Chinese proverb goes, "You can't have the best of both worlds." Fortunately, remote sensing spatial–temporal–spectral fusion (STSF) can break through the inherent limitations of the sensor hardware imaging system to obtain high spatial–temporal–spectral resolution images.

In recent decades, extensive image fusion methods for remote sensing have been put forward and most of them could be basically classified into three categories: 1) spatial–spectral fusion; 2) spatial–temporal fusion; and 3) STSF.

Spatial–spectral fusion is intended to generate images with both high spatial and spectral resolutions [7], [8], [9], [10], [11], such as the fusion of the high-spatial-resolution (HR) panchromatic (PAN) image and the low-spatial-resolution (LR) multispectral (MS) image, termed as pansharpening [12]. The typical algorithms can be classified into four categories [13]: 1) component substitution (CS)-based methods; 2) multiresolution analysis (MRA)-based methods; 3) variational optimization (VO)-based methods; and 4) learning-based methods. CS-based methods decompose the LR-MS into spatial components and color spaces, where the former is replaced by the HR-PAN image, such as principal component analysis (PCA) [14] and Gram–Schmidt (GS) [15]. MRA-based methods project PAN and MS images into scale levels with multiresolution filters and spatial detail from PAN is then injected into MS, such as adaptive wavelet (AW) [16] and generalized Laplacian pyramid (GLP) [17]. However, CS-based methods usually expose defects in spectral structure preservation; in contrast, MRA-based methods are conservative in spatial detail enhancement [18]. VO-based methods regard the solution of the image fusion as an ill-posed inverse problem and build a degradation model between the observations and ideal image for optimization [19]. Despite providing acceptable results, the VO-based methods are confined to relatively lower computing efficiency due to the iterative optimization process. Recently, learning-based methods have achieved competitive power in computer vision with remarkable nonlinear abilities, especially deep learning (DL) [20], [21], [22], [23]. The PNN was the first DL-based pansharpening method on a simple three-layer convolutional neural network (CNN) [24]. The PANNet carried out a

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feature process in the high-frequency domain to keep spatial structure [25]. Subsequently, Liu et al. [26] proposed an interesting two-stream fusion network (TFNet) with a much deeper structure for data training. The memory-augmented deep conditional unfolding network (MDCUN) with good interpretability attempted to capture the long-range similarities between PAN and MS patches to facilitate texture enhancement [7]. In addition, the MS and hyperspectral (HS) fusion method, mainly served as the extension of pansharpening, have attracted ever-increasing attention, such as coupled nonnegative matrix factorization (CNMF) [27] and HS image super-resolution (Hysure) method [28]. Moreover, the DL-based MS/HS fusion methods are also springing up due to their amazing performance, for example, the multiscale low-rank deep back-projection fusion network (MLR-DBPFN) [29] and the dual-cross Transformer-based fusion (DCTransformer) for MS/HS fusion [30].

Spatial–temporal fusion is intended to generate images with high spatial and temporal resolutions [31], [32], [33], [34], [35]. Most methods have been developed on Landsat and MODIS data to acquire the daily HR data with 30-m spatial resolution [36]. The spatial–temporal fusion methods can be roughly divided into four categories [37]: 1) unmixing-based methods; 2) weight function-based methods; 3) Bayesian-based methods; and 4) learning-based methods. Unmixing-based methods estimate the fine pixel from the spectrum of the coarse pixel based on the linear spectral mixture model (LSMM), such as the spatial–temporal data fusion approach (STDFA) [38]. Weight function-based methods transfer the temporal change on the coarse pixel to the fine pixel using weighted functions, while the pure coarse pixel in the neighborhood is given a higher weight. The spatial and temporal adaptive reflectance fusion model (STARFM) [39] and enhanced STARFM (ESTARFM) [40] are the typical methods. Bayesian-based methods introduce maximum a posterior (MAP) framework and establish the relationship between the observations and ideal image, such as Bayesian maximum entropy (BME) [41]. However, most above spatial–temporal fusion methods show limitations in heterogeneous landscapes. As mentioned above, learning-based methods model the nonlinear spatial–temporal reliance between coarse and fine pixels. For instance, the sparse-representation-based spatiotemporal reflectance fusion model (SPSTFM) exploited the relation of difference map between HR and LR images to rebuild the details of change [42]. Similarly, DL has emerged in spatial–temporal fusion and shown strong capabilities in spectral fidelity and land-cover change prediction, such as a two-stream CNN for spatiotemporal image fusion (StfNet) [43] and a multistream fusion network (MSNet) [44].

Nevertheless, most exploration of image fusion has lingered on synthesizing two kinds of indicators among spatial, temporal, and spectral resolutions, as shown in Fig. 1. Therefore, an integrated STSF framework is urgently required to comprehensively utilize multisource and multitemporal data to produce images with high spatial, temporal, and spectral resolutions.

To date, some breakthroughs have been made in STSF, and most studies can be concluded into two categories:

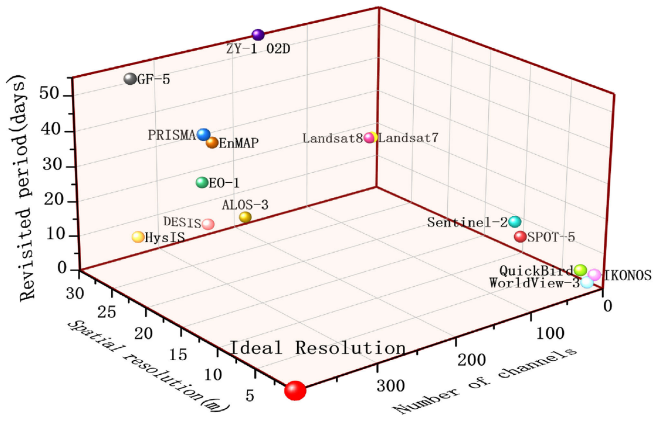


Fig. 1. There is a fundamental tradeoff among the temporal, spatial, and spectral resolutions for most current remote sensing satellite observations.

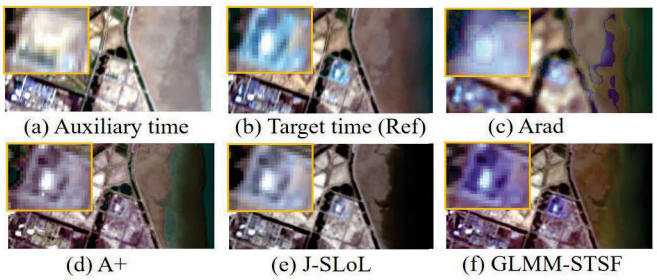


Fig. 2. Existing methods show spectral distortion in complex situations with land-cover changes. (a) and (b) MS images with land-cover changes on the auxiliary and target time, respectively. (c)–(f) Fused results by existing different methods.

1) stepwise-based STSF and 2) integrated STSF. For the former, Chen et al. [45] decomposed the STSF into stepwise processing of spatial–spectral fusion and spatial–temporal fusion. Zhao and Huang [46] designed successive subtasks by the order of visual band sharpening, spatial–temporal fusion, and temporal–spectral fusion. For the latter, Shen [47] first proposed an integrated STSF based on a MAP framework. Huang et al. [48] explored the relevance of different fusion modes and produced the result using a general function. Shen et al. [49] developed a unified sparsity model to thoroughly fuse an arbitrary number of multisource and multitemporal satellite data. Peng et al. [50] adopted a sparse tensor model to depict the spatial self-similarity and spectral redundancy in HS. Zhou et al. [51] proposed a novel generalized linear spectral mixing model, which transferred the linear spectral change to the domain of the endmember and abundance matrix of the HS image for estimation. Recently, DL-based methods have been proposed to model complex nonlinear mapping relationships in STSF. Chen et al. [52] proposed a progressive STSF network (PSSTFN), which integrated spatial–spectral fusion and spectral–temporal fusion into a unified end-to-end fusion framework. Liu et al. [53] proposed an unsupervised spatial–temporal–spectral fusion model (USTSFM) with dual-task interactive learning, which achieved accurate prediction on complex land-cover changes by the modeling and interaction of temporal and spectral relationship branches.

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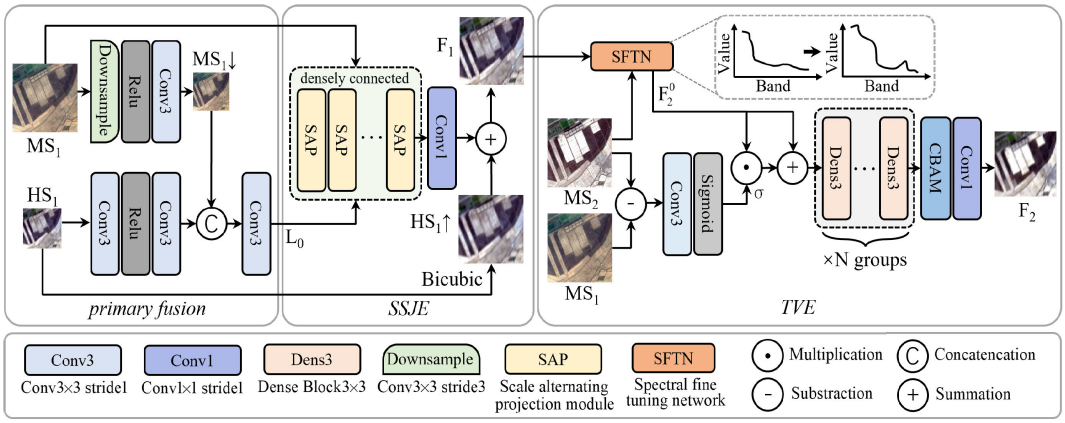


Fig. 3. Flowchart of the proposed DC-STSF. From left to right, the SSJE subnet and the TVE subnet are shown, respectively.

However, there are issues to be solved. On the one hand, most studies [47], [48], [49] assumed that LR-HS images (e.g., MODIS) have a much shorter revisit period than HR-MS images (e.g., Landsat), which conforms with the characteristics of most space-borne HS data (e.g., ZY1-02D and GaoFen-5). On the other hand, the prediction of changes on heterogeneous land cover is challenging, and most current STSF methods performed poorly in addressing nonlinear temporal changes, as shown in Fig. 2. In this article, we proposed a dual-task cascaded network for STSF (DC-STSF), and the main contributions are as follows.

1) We propose a DC-STSF network, by cascading spatial–spectral joint enhancement (SSJE) and temporal variation estimation (TVE) into a unified framework.
2) The SSJE task with an iterative alternating projection module is designed. Furthermore, the spatial enhancement unit (SEU) and the error correction unit (ECU) on feedback fusion of errors are developed to enhance spatial and spectral fidelity.
3) The TVE on the spectral fine-tuning network (SFTN) is developed, by considering the inconsistent spectral overlap range between HR-MS and LR-HS images, to further enhance the temporal and spectral fidelity.

The rest of the article is organized as follows. In Section II, the main architecture of the network is analyzed. In Section III, we conducted the experiments on reduced- and full-resolution data and compared the proposed method with previous methods qualitatively and quantitatively. In Section IV, we summarized the contents and made a brief prospect of future work.

II. Proposed Method

The $\mathrm{MS}_1 \in \mathbb{R}^{H\times W\times b}$ and $\mathrm{HS}_1 \in \mathbb{R}^{h\times w\times B}$ refer to the HR-MS and LR-HS images at auxiliary time t_1, respectively, where h and H or w and W refer to the dimensions of rows or columns, B and b denote spectral bands ($H \gg h$, $W \gg w$, and $B \gg b$). $\mathrm{MS}_2 \in \mathbb{R}^{H\times W\times b}$ and $\mathrm{HS}_2 \in \mathbb{R}^{h\times w\times B}$ refer to the HR-MS and LR-HS images at target time t_2. $F_1 \in \mathbb{R}^{H\times W\times B}$ and $F_2 \in \mathbb{R}^{H\times W\times B}$ are the results of the SSJE and TVE tasks, respectively, while the corresponding ground truths are $X_1 \in \mathbb{R}^{H\times W\times B}$ and $X_2 \in \mathbb{R}^{H\times W\times B}$.

A. General Framework

The STSF framework is formed with two tasks that cascade in succession, and the SSJE is first completed to produce F_1 with high spatial–spectral resolution, and then the TVE model is performed to generate F_2 with high temporal–spatial–spectral resolution. This process can be simplified as follows:

$$F_1 = \mathrm{SSJE}(\mathrm{MS}_1, \mathrm{HS}_1) \tag{1}$$

$$F_2 = \mathrm{TVE}(\mathrm{MS}_1, \mathrm{MS}_2, F_1) \tag{2}$$

where the tasks of SSJE$(\cdot)$ and TVE$(\cdot)$ are trained in an end-to-end network as shown in Fig. 3.

B. SSJE Modeling Task

The general operation to deal with the scale difference between HR-MS and LR-HS images is to resample the data at the initial stage of the network. Referring to the ideas of deep back-projection networks [54], we realize iterative cross-scale fusion by a scale alternating projection (SAP) for gaining consensus on scale conversion and preserving the spatial–spectral structure.

The MS_1 is first downsampled to the size of HS_1 through a convolution layer with a step size of 3 and then projected to the dimension of $h \times w \times B$. The HS_1 undergoes similar operations as follows:

$$F_m = \mathrm{Cov}_3(\mathrm{relu}(d(\mathrm{MS}_1))) \tag{3}$$

$$F_h = \mathrm{Cov}_3(\mathrm{relu}(\mathrm{Cov}_3(\mathrm{HS}_1))) \tag{4}$$

where $d(\cdot)$ denotes the downscaling operator and $\mathrm{Cov}_3(\cdot)$ denotes the convolution layer with a step size of 3. relu$(\cdot)$ is a rectified linear module. F_h and F_m are then concatenated to obtain primary fusion feature L_0 as follows:

$$L_0 = \mathrm{Conv}_3(\mathrm{Concat}(F_h, F_m)) \tag{5}$$

where Concat$(\cdot)$ denotes channel-wise feature concatenation.

Second, we executed an SAP module with the SEU and the ECU, which undertakes to inject HR-MS spatial detail into the feature domain in forward propagation and provides fusion error feedback in backward propagation. In general, taking the nth layer as an example, the specific architecture of the SAP is shown in Fig. 4.

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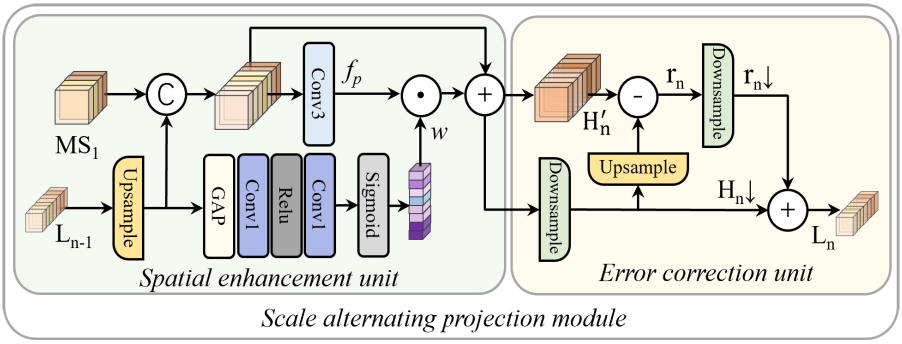


Fig. 4. Flowchart of the SAP module with internal SEU and ECU.

The spatial enhancement unit is proposed to inject the details extracted from the HR-MS image into the feature domain. The primary fusion feature L_0 is upsampled and concatenated with MS_1 to obtain the synthesized feature f_p, which is defined as follows:

$$f_p = \mathrm{Conv}_3(\mathrm{Concat}(\mathrm{MS}_1, u(L_{n-1}))) \tag{6}$$

where $u(\cdot)$ denotes the upsampling operator. In addition, the channel attention module is explored to model the spectral characteristics, which are depicted as follows:

$$w = \sigma(f_{c2}(\mathrm{relu}(f_{c1}(\mathrm{gap}(L_{n-1}))))) \tag{7}$$

$$H_n = w \cdot (f_p + \mathrm{Concat}(\mathrm{MS}_1, u(L_{n-1}))) \tag{8}$$

where $w = \{w_1, w_2, \ldots, w_B\}$, $w_i \in [0, 1]$, means the weight of each channel. $\mathrm{gap}(\cdot)$ denotes the global average pooling (GAP) layer. $f_{c1}(\cdot)$ and $f_{c2}(\cdot)$ denote the fully connected layers. $\sigma(\cdot)$ denotes sigmoid activation.

Finally, the original spatial content from MS_1 is explored as a supplement to the fusion process, and spectral information is reweighted considering the channel characteristics by the weighting vector w. Moreover, the output of the SEU from the previous stages is further explored to ensure the stable propagation of the gradient and lower parameters as

$$H'_n = H_n + H_{n-1} + \cdots + H_1. \tag{9}$$

The output H_n is then fed into the ECU.

The ECU is proposed on error feedback to realize feature self-correcting in the backward propagation. Concretely, H_n is downsampled to gain an LR version $H_n \downarrow$ and then upsampled to obtain a blurred version $H_n \downarrow\uparrow$ with the same size. The error with high-frequency texture details is obtained by subtracting H_n and $H_n \downarrow\uparrow$. Finally, r_n is downsampled and feedback it to the $H_n \downarrow$ to create the optimized result L_n, as

$$H_n \downarrow = d\left(H'_n\right) \tag{10}$$

$$H_n \downarrow\uparrow = u(\mathrm{relu}(H_n \downarrow)) \tag{11}$$

$$r_n = H'_n - H_n \downarrow\uparrow \tag{12}$$

$$L_n = H_n \downarrow + d(r_n) \tag{13}$$

where $H_n \downarrow\uparrow$ and $H_n \downarrow$ mean resampled and downsampled feature, respectively, and r_n refers to the residual error. Similarly, L_n is updated through dense connection as follows:

$$L'_n = L_n + L_{n-1} + \cdots + L_1. \tag{14}$$

The ECU projects features from the HR domain into the LR domain and promotes the fusion effect through the cognition of scale relation between LR-HS and HR fused images.

The SAP module creates L_n after n repeated iterations, then we only need to execute separate the SEU to get an HR H_{n+1}. The HR feature is connected with interpolated image $\mathrm{HS}_1 \uparrow$ as a residual so that the result obtains a slight spectral deviation

$$F_1 = \mathrm{Conv}_1(H_{n+1}) + \mathrm{HS}_1 \uparrow \tag{15}$$

where F_1 is the result of SSJE and $\mathrm{HS}_1 \uparrow$ is obtained by bicubic interpolation.

C. TVE Modeling Task

The temporal variation exhibits irregularity on heterogeneous surfaces. The primary fusion result F_1, that is, the HR-HS on the auxiliary phase, generated by the SSJE task is considered as a spectral prior for the TVE.

The SFTN is designed by considering the difference in the spectral coverage between the MS image MS_2 of the target phase and F_1. The SFTN is arranged to apply finite spectral information from MS_2 to adaptively modulate the spectral vector distribution of F_1, the architecture of the SFTN is shown in Fig. 5. The core concept of the SFTN is simplified as follows:

$$F_p^m = f_1\left(F_1^m\right) \tag{16}$$

$$F_p^{um} = f_2\left(F_1^{um}\right) + g_1\left(F_p^m\right) \tag{17}$$

where F_1^m and F_1^{um} refer to the spectral bands within and outside the overlap, respectively. F_p^m and F_p^{um} are corresponding prediction results. $f_1(\cdot)$ and $f_2(\cdot)$ refer to nonlinear temporal transform, and $g_1(\cdot)$ refers to the band-inferred function. For simplicity, the above equations can be deformed into

$$F_p = \mathrm{Concat}\left(f(F_1), g\left(F_1^m\right)\right) \tag{18}$$

where $f(\cdot)$ and $g(\cdot)$ refer to nonlinear temporal transform. F_p can supply a rough spectral curve. The computed result of transformation (18) is equivalent to (16) and (17).

Although only MS_2 affords the spectral reference, it is conceivable that the temporal–spectral change of each pixel involves a potential logicality, which can be quantified to channel-wise parameters α and offset β. Consequently, a weight generator is designed with residual blocks and convolutional block attention modules (CBAMs) to learn deep-level

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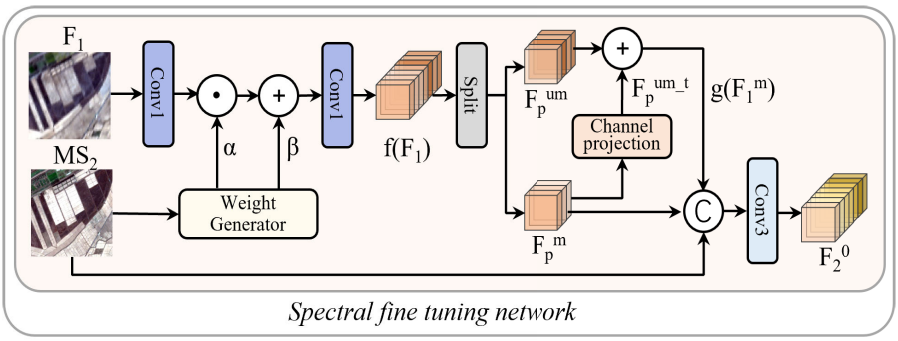


Fig. 5. Flowchart of the SFTN.

spectral knowledge, and the generated feature is divided into two parts along the channel dimension. We use a GAP layer to extract α and β from the feature domain, and the process is depicted as follows:

$$\varphi = \mathrm{CBAM}_2(\mathrm{res}_2(\mathrm{CBAM}_1(\mathrm{res}_1(\mathrm{MS}_2)))) \tag{19}$$

$$\alpha, \beta = \mathrm{gap}(\rho(\varphi)) \tag{20}$$

where res_1 and res_2 denote residual blocks, φ and ρ denote generated feature maps and channel-wise split, respectively,

$$f(F_1) = \mathrm{Conv}_1((\alpha + 1) \cdot \mathrm{Conv}_1(F_1) + \beta) \tag{21}$$

where $f(F_1)$ denotes the temporal transform result. The front and rear 1×1 convolution layers are used to represent channel expansion and compression. Therefore, $f(F_1)$ is divided into two sections as ${F_p}^m$ and ${F_p}^{um}$ in accordance with the coverage to infer nonoverlapping content using the correlation among the bands. To acquire ${F_p}^{um}$, ${F_p}^m$ simply adopts a gate to filter out valuable channel-wise messages for expansion. Then, the feature is projected to the interval of ${F_p}^{um}$ to obtain ${F_p}^{um_t}$ and $g(F_1^m)$

$${F_p}^{um_t} = \mathrm{Conv}_1\left({F_p}^m \cdot \sigma\left(\mathrm{Conv}_3\left({F_p}^m\right)\right) + {F_p}^m\right) \tag{22}$$

$$g\left(F_1^m\right) = {F_p}^{um} + {F_p}^{um_t} \tag{23}$$

where ${F_p}^{um_t}$ is the mapped feature. Subsequently, the fine-tuned results are cascaded with MS_2 as a preliminary fusion of spatial and spectral information, which is depicted as follows:

$$\begin{aligned} {F_2}^0 &= \mathrm{Conv}_3(\mathrm{Concat}(F_p, \mathrm{MS}_2)) \\ &= \mathrm{Conv}_3\left(\mathrm{Concat}\left(f(F_1), g\left(F_1^m\right), \mathrm{MS}_2\right)\right) \end{aligned} \tag{24}$$

where ${F_2}^0$ denotes the preliminary fusion feature from the SFTN. The difference image (DI) is explored to highlight the variable regions to further refine the details of the spectrum. In addition, to avoid the artifacts at the edges, the sigmoid function is used to suppress the low correlation noise in DI. It is depicted as follows:

$$s = \mathrm{Conv}_3(\sigma(\mathrm{MS}_2 - \mathrm{MS}_1)) \tag{25}$$

$$F_2^1 = {F_2}^0 \cdot s + {F_2}^0 \tag{26}$$

where s is generated for guidance. F_2^1 shows more concern for variations through weight allocation. The integrated image is then projected into the deep feature domain by three sets of densely connected convolutional blocks. It is depicted as follows:

$$F_2^{1,i} = \mathrm{Dens}_i\left(F_2^1\right) \tag{27}$$

where $i = \{1, 2, 3\}$. $\mathrm{Des}_i(\cdot)$ is densely connected convolutional blocks with a kernel size of 3. To better rebuild fine images from semantic features, the CBAM is further adopted for interested region enhancement

$$F_2 = \mathrm{Conv}_1\left(\mathrm{CBAM}\left(F_2^{1,i}\right)\right) \tag{28}$$

where F_2 is the ultimate fusion result.

D. Loss Function

The DC-STSF is trained with the joint loss function composed of general loss L_{MAE1} and L_{MAE2}, spatial gradient loss L_{grad} and spectral angle mapper (SAM) loss L_{SAM} for synthetical constrain, and the weight of each item depends on previous summary and test as follows:

$$\mathrm{Loss} = \lambda_1 * L_{\mathrm{MAE1}} + \lambda_2 * L_{\mathrm{MAE2}} + \lambda_3 * L_{\mathrm{grad}} + \lambda_4 * L_{\mathrm{SAM}}. \tag{29}$$

General reconstruction loss measures the mean absolute error of SSJE and TVE tasks, which encourages F_2 (or F_1) as close as possible to ground truth X_2 (or X_1). The formula is depicted as follows:

$$L_{\mathrm{MAE1}} = \| X_1 - F_1 \|_1 \tag{30}$$

$$L_{\mathrm{MAE2}} = \| X_2 - F_2 \|_1 . \tag{31}$$

Spatial gradient loss determines the spatial consistency of F_2 and ground truth X_2 in the high-frequency domain with the spatial edge extracted by the Laplacian operator. The formula is depicted as follows:

$$L_{\mathrm{grad}} = \| g(X_2) - g(F_2) \|_2 \tag{32}$$

where $g(\cdot)$ denotes the 3×3 Laplacian operator.

SAM loss is used to guarantee the color authenticity and spectral structure of F_2 in spectral dimension. The formula is depicted as follows:

$$L_{\mathrm{SAM}} = \frac{1}{HW} \sum_{h,w} \arccos\left(\frac{\tilde{X}_{h,w} \cdot \tilde{F}_{h,w}}{\left\|\tilde{X}_{h,w}\right\|_2 \left\|\tilde{F}_{h,w}\right\|_2} \right) \tag{33}$$

where H and W denote the dimensions of rows and columns, respectively. $\tilde{X}_{h,w}$ and $\tilde{F}_{h,w}$ denote spectral vector at pixel (h, w).

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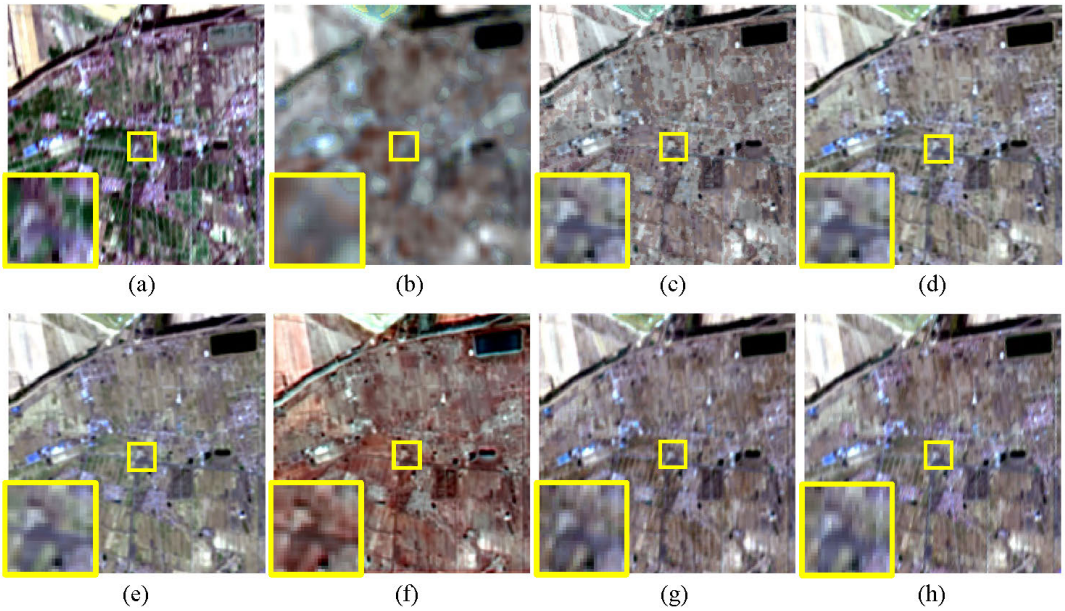


Fig. 6. Reduced-resolution results on the DY dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

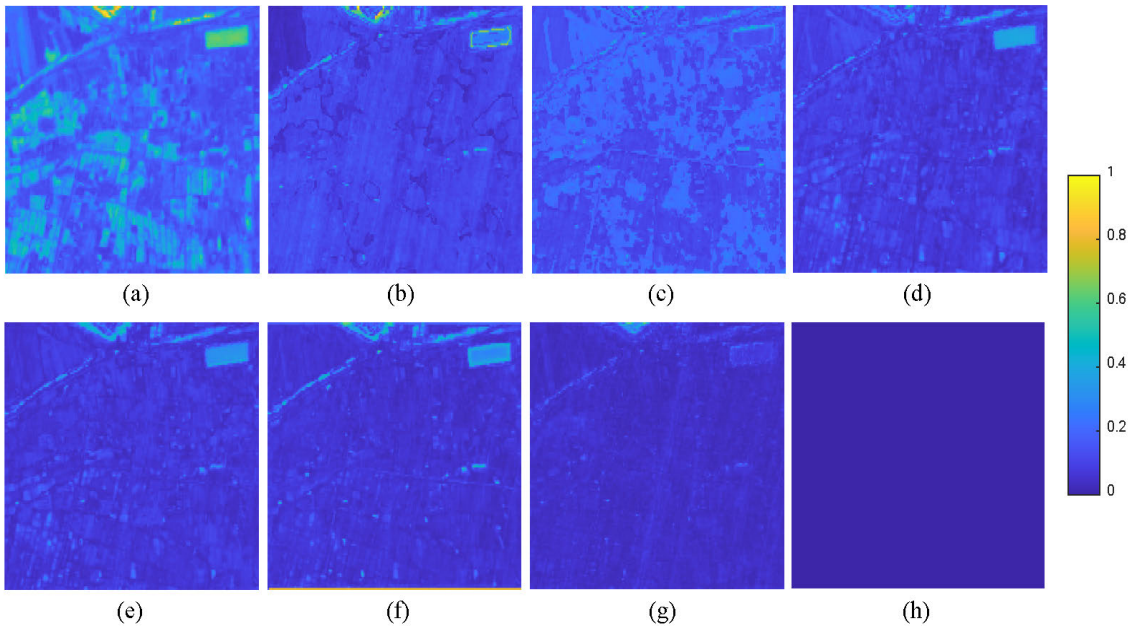


Fig. 7. SAM error images corresponding to Fig. 6 on the DY dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

III. Experimental Results

A. Dataset and Experimental Settings

Sentinel-2 MS and ZY1-02D HS were applied to evaluate the performance. The HR-MS image with R, G, B, and NIR spectral bands from Sentinel-2 features by 10-m spatial resolution. ZY1-02D provided an LR-HS image with a 30-m spatial resolution, and we manually removed 15 bad bands from the source images.

For training data, the original images were spatially degraded with the Gaussian blur kernel and the downsampling operator following Wald's protocol, and the original LR-HS images were converted to the ground truth. The MS and HS images were cropped into patches with spatial dimensions of 72×72 and 24×24, respectively. Accordingly, we gathered about 4500 pairs of MS-HS patches for the Dongying (DY) dataset and nearly 2000 pairs for the Yellow River Estuary (YRE) dataset, and the details are listed in Table I.

The network was trained and tested on an NVIDIA GeForce RTX 3090 GPU and Intel i7-11700K CPU. We initialize the batch size to 16 and the learning rate to $1 \times e^{-4}$ with 0.5 decay per 25 epochs. λ_1, λ_2, λ_3, and λ_4 are set to 0.2, 1, 0.0001, and 0.001, respectively. In addition, to prevent gradient explosion, gradient clipping was performed in the

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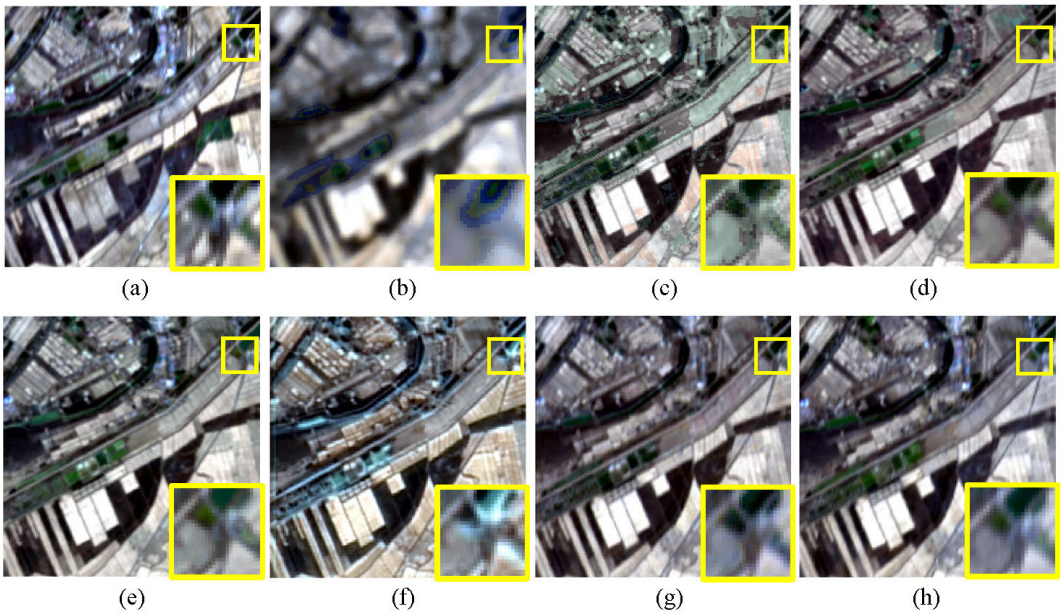


Fig. 8. Reduced-resolution results on the YRE dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

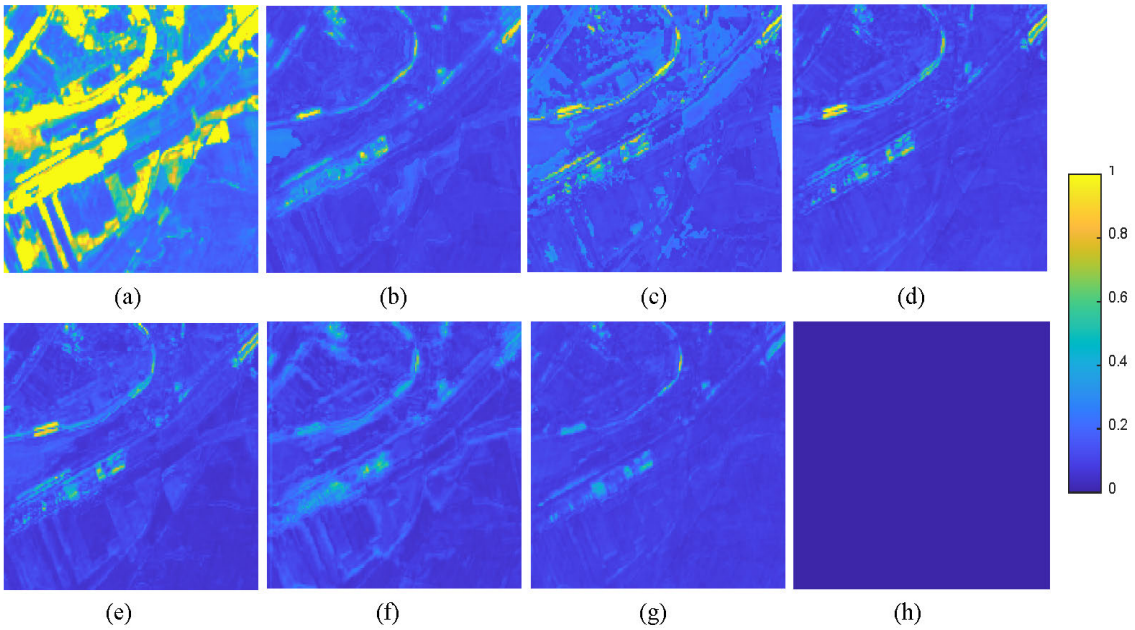


Fig. 9. SAM error images corresponding to Fig. 8 on the YRE dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

training. When approaching 150 epochs, a convergent model was obtained.

B. Comparative Methods and Metrics

Due to the absence of target time HSI in the fusion model construction, most currently available integrated STSF methods cannot be directly applied. To verify the superiority of DC-STSF, five common methods were used for comparison, including three spectral super-resolution methods of Arad [55], A+ [56], and J-SLoL [57] and two STSF methods of GLMM-STSF [51] and USTSFM [53]. The selection of all methods is considered applicable and has been fairly compared with respect to the training strategy and parameter settings of the original paper.

The fusion accuracies refer to six common metrics, including the peak signal-to-noise ratio (PSNR) [58], root mean square error (RMSE) [59], erreur relative globale adimensionnelle de synthèse (ERGAS) [60], structural similarity (SSIM) [61], SAM [62], and spatial correlation coefficient (SCC) [63]. The PSNR, RMSE, ERGAS, and SSIM are metrics to measure global performance. SAM and SCC were adopted to measure the spectral and spatial quality, respectively [64].

The PSNR is used to represent the signal reconstruction quality of the fused image in each band, and the average value

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TABLE I
SPECIFIC INFORMATION OF THE DY DATASET AND THE YRE DATASET

Dataset	Settings	Sentinel-2 (MS)	ZY1-02D (HS)
DY	Auxiliary time t_1	December 8, 2019	December 11, 2019
	Target time t_2	February 10, 2021	February 10, 2021
	Spatial resolution (m)	10	30
	Bands number	Original: 10 / remain: 4	Original: 166 / remain: 151
	Training data	72×72×4000 patches	24×24×4000 patches
YRE	Auxiliary time t_1	December 8, 2019	December 11, 2019
	Target time t_2	February 10, 2021	February 10, 2021
	Spatial resolution (m)	10	30
	Bands number	Original: 10 / remain: 4	Original: 166 / remain: 151
	Training data	72×72×1500 patches	24×24×1500 patches

TABLE II
QUANTITATIVE RESULTS ON REDUCED-RESOLUTION DATASETS. (THE BEST IS MARKED IN BOLD LETTERS AND THE SECOND-BEST IS UNDERLINED)

Dataset	Method	PSNR	RMSE	ERGAS	SSIM	SAM	SCC
DY	Arad	24.4734	0.0321	4.8089	0.7401	6.8957	0.5759
	A+	19.3202	0.0633	12.5311	0.7355	9.6038	0.6704
	J-SLoL	25.3062	0.0268	4.1516	0.8079	5.5230	0.7692
	GLMM-STSF	23.3474	0.0303	4.6982	0.8305	6.4628	0.7824
	USTSFM	25.6297	0.0288	4.6993	0.8451	5.4814	0.7693
	DC-STSF(ours)	**30.4601**	**0.0184**	**2.7436**	**0.9101**	**3.9915**	**0.8754**
YRE	Arad	22.9025	0.0307	5.7279	0.7286	5.7279	0.5956
	A+	16.3973	0.0792	9.4922	0.5355	9.9993	0.4359
	J-SLoL	20.6348	0.0400	4.6075	0.6537	4.7918	0.5355
	GLMM-STSF	19.5311	0.0408	6.1455	0.6664	5.4767	0.5706
	USTSFM	21.4067	0.0343	5.1508	0.6806	4.1069	0.5594
	DC-STSF(ours)	**27.8993**	**0.0170**	**2.4709**	**0.8531**	**2.9459**	**0.8297**

serves as an overall assessment. The formula is as follows:

$$\mathrm{PSNR} = \frac{1}{B}\sum_{b=1}^{B} 20\log_{10}\frac{\max\left(X_2^b\right)}{\sqrt{\mathrm{mse}\left(X_2^b, F_2^b\right)}} \tag{34}$$

where F_2^b and X_2^b represent the bth band of the fused image F_2 and the ground truth X_2, respectively. MSE indicates the mean square error function, and B represents the total number of bands.

RMSE indicates the difference in spectral and spatial dimensions between the fused image and the ground truth. The formula is as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{HWB}\sum_{h,w,b}\left(F_2^b(h,w) - X_2^b(h,w)\right)^2} \tag{35}$$

where H and W represent the number of pixels in rows and columns, respectively. $F_2^b(h,w)$ and $X_2^b(h,w)$ denote the pixel (h,w) value of the fused image in the bth band, respectively. The ideal value of RMSE is 0.

ERGAS comprehensively reflects the spatial and spectral qualities of F_2. The formula is as follows:

$$\mathrm{ERGAS} = 100\frac{h}{o}\sqrt{\sum_{b=1}^{B}\left(\frac{\mathrm{RMSE}_b}{\left(\bar{X}_2^{h,w}\right)^2}\right)^2} \tag{36}$$

where h/o refers to the scale ratio of LR-HS and HR-MS images. RMSE_b represents the RMSE of the bth band. The ideal value of ERGAS is 0.

The SSIM is used to reflect the SSIM between the fused images and the ground truth, measuring the degree of distortion from three dimensions: brightness, contrast, and structure. The formula is as follows:

$$\mathrm{SSIM} = \frac{\left(2\mu_x\mu_f + c_1\right)}{\left(\mu_x^2 + \mu_f^2 + c_1\right)}\frac{\left(2\sigma_{xf} + c_2\right)}{\left(\sigma_x^2 + \sigma_f^2 + c_2\right)} \tag{37}$$

where μ_x and μ_f represent the mean value of the fused image and the ground truth, respectively. σ_x and σ_f represent the standard deviation. σ_{xf} denotes the covariance between images. c_1 and c_2 denote constants. The ideal value of SSIM is 1.

The SAM indicates the spectral quality of F_2. The formula is as follows:

$$\mathrm{SAM} = \frac{1}{HW}\sum_{h,w}\arccos\left(\frac{\left\langle \hat{X}_2^{h,w}, \hat{F}_2^{h,w}\right\rangle}{\left\|\hat{X}_2^{h,w}\right\|_2\left\|\hat{F}_2^{h,w}\right\|_2}\right) \tag{38}$$

where $\hat{F}_2^{h,w}$ and $\hat{X}_2^{h,w}$ represent the spectral vector of F_2 and X_2 at the pixel (h,w), respectively. $\langle\cdot,\cdot\rangle$ denotes the vector inner product. The ideal value of SAM is 0.

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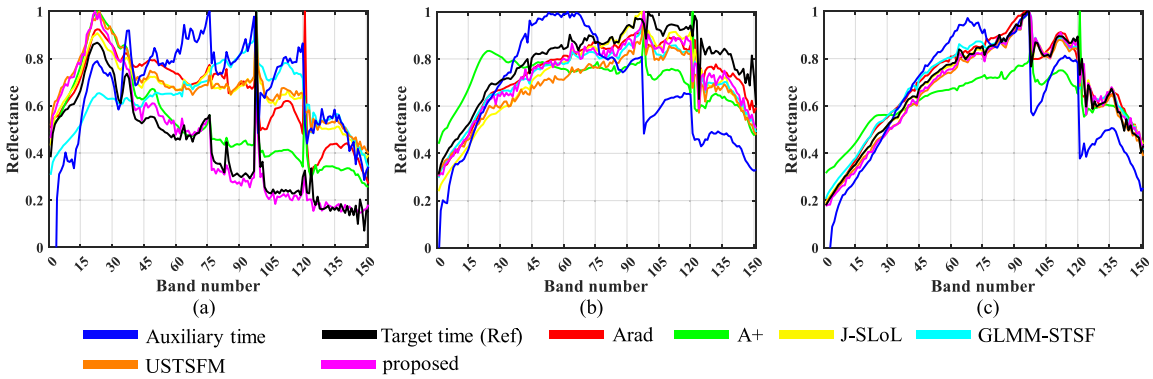


Fig. 10. Reflectance curve comparison of typical surface features for different results corresponding to Fig. 6 on the DY dataset. (a) Water. (b) Vegetation. (c) Buildings.

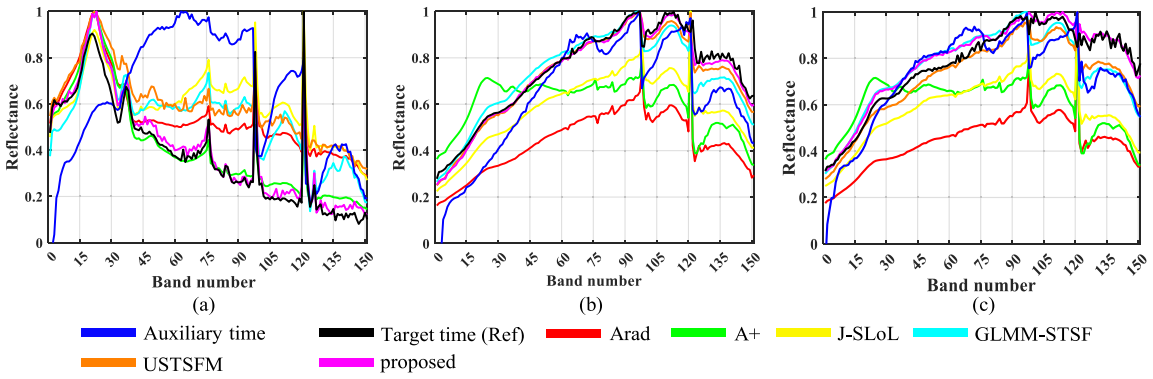


Fig. 11. Reflectance curve comparison of typical surface features for different results corresponding to Fig. 7 on the YRE dataset. (a) Water. (b) Vegetation. (c) Buildings.

The SCC is used to measure the spatial performance by averaging the SCC value of each band. The formula is as follows:

$$\mathrm{SCC} = \frac{1}{B}\sum_{b=1}^{B}\frac{\sum_{h,w}\left(F_2^b-\overline{F_2^b}\right)\left(X_2^b-\overline{X_2^b}\right)}{\sqrt{\sum_{h,w}\left(F_2^b-\overline{F_2^b}\right)^2\sum_{h,w}\left(X_2^b-\overline{X_2^b}\right)^2}}. \quad (39)$$

The ideal value of SCC is 1.

C. Experimental Results on Reduced-Resolution Datasets

1) Quantitative Comparison: The network was trained on reduced-resolution data obtained from DY and YRE datasets. The quantitative results of the two datasets are shown in Table II, where the best values are marked with bold letters and the second-best is underlined. In Table II, as can be seen, the spatial quality of Arad and PSNR of A+ is inadequate, while J-SLoL, GLMM-STSF, and USTSFM performed relatively close indicators. Our proposed method achieves state-of-the-art results on the DY dataset, especially in the PSNR and SCC. The quantitative comparison in the YRE dataset is shown in the second row of Table II. Our proposed method achieves the best performance in all the quality indices, which means that our method achieves the best spatial and spectral qualities. The indicators in A+ are poor, while Arad produces relatively competitive results. The gaps between J-SLoL, GLMM-STSF, and USTSFM are negligible, and the spectral fidelity of these methods is better than Arad.

2) Qualitative Comparison: The intuitive visual comparisons are shown in Fig. 6 corresponding to the DY dataset. The bands R, G, and B (refer to bands 29, 19, and 10) from fused HR-HS images are chosen to synthesize pseudo-color images for display. Fig. 6(a) and (h) refers to original images in adjacent time and target time, which is conducive to displaying dynamic land-cover and spectral changes, respectively. The contrast among Fig. 6(b)–(g) demonstrates that Arad suffers from severe spatial blurring, while A+ and USTSFM are accompanied by obvious color aberration. In the vegetation area, J-SLoL and GLMM-STSF exhibit obvious spectral distortion. It is observed that the result of DC-STSF is more approximate to the reference from the perspective of space and spectrum. Compared to other STSF methods, DC-STSF accurately predicted the violent and irregular spectral variation during the period. In terms of the zoomed image in the lower left corner, it effectively verified that DC-STSF performs better than other methods.

For ease of perceiving the subtle differences among competitors, we further added the SAM residuals between the reference and the results to highlight the confused areas in Fig. 7. Among them, it is noticed that our method is closer to the reference than other comparison methods, which also accords with the visualization and metrics.

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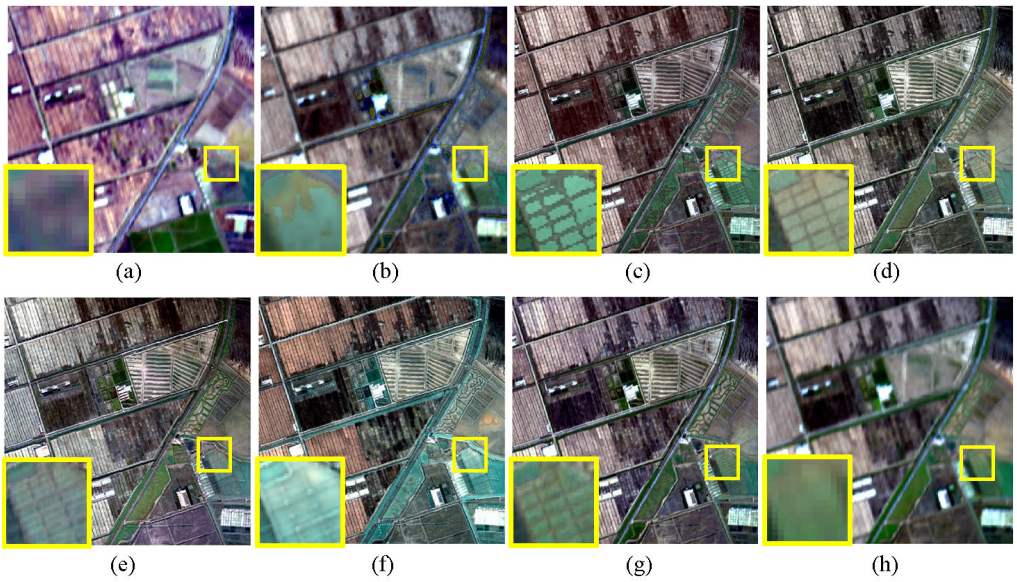


Fig. 12. Full-resolution results on the DY dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

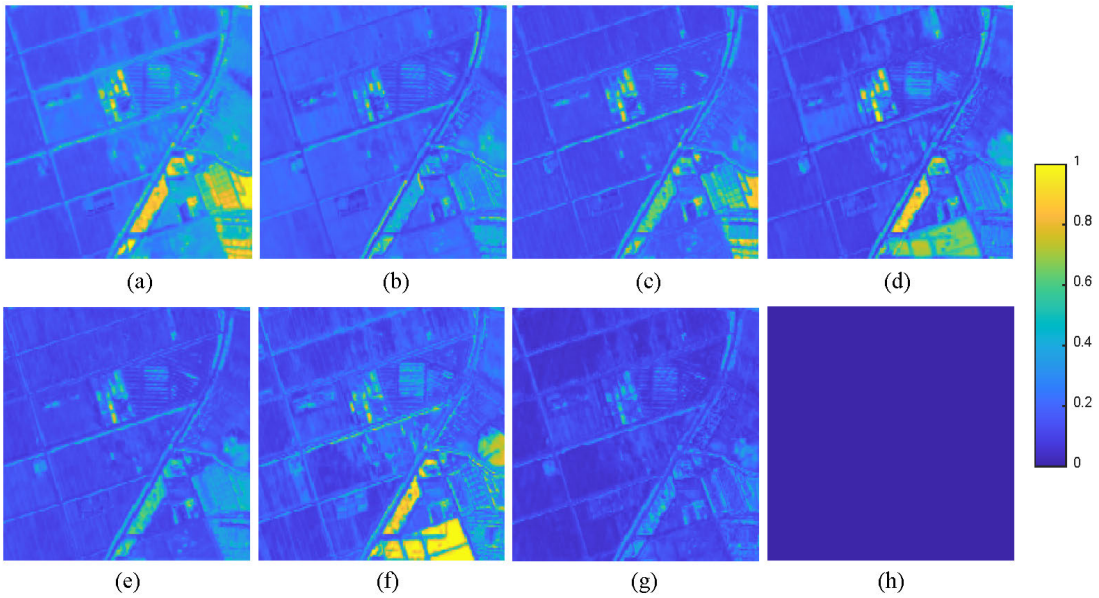


Fig. 13. SAM error images corresponding to Fig. 12 on the DY dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

In Fig. 8, the intuitive visual comparisons corresponding to the YRE dataset are shown, where Arad, J-SLoL, and DC-STSF still perform relatively delicate spectrum while the rest causes apparent distortions. To some extent, the former has proved its better robustness, while DC-STSF not only realizes the cross-temporal spectral fidelity but also effectively exploits the spectral information of adjacent time as an assistant. However, Arad has never been subject to the problem of spatial ambiguity. In Fig. 9, the proposed method still keeps the minimum residual amount, that is, it has well eliminated the unstable region highlighted in Fig. 9(a). Furthermore, to visualize the recovery precision of each independent band of the comparison methods in DY and YRE datasets, we depict the reflectance curves of specific pixels belonging to three typical landscapes, including water, vegetation, and buildings. As shown in Figs. 10 and 11, Arad and A+ deviates from the standard curve evidently, USTSFM has a good degree of fitting, but it shows unstable in the water landscape. The above results are further proof of the superior abilities of our method, while the proposed method reflects good band correlation and robustness to land covers.

D. Experimental Results on Full-Resolution Datasets

To ascertain the application value of the methods in the real-world data, we enriched with the full-resolution experiments under the original datasets in Figs. 12–14.

1) Qualitative Comparison: Since there is no suitable indicator to evaluate the fusion results, only qualitative evaluation

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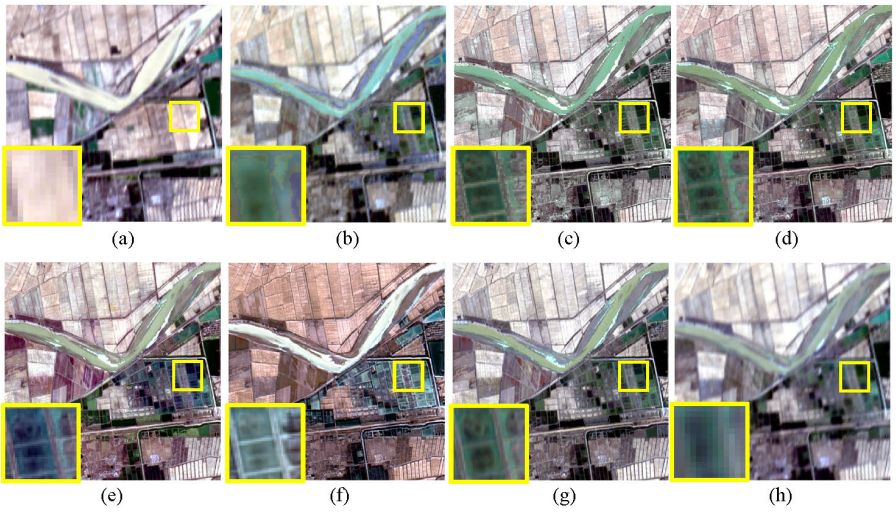


Fig. 14. Full-resolution results on the YRE dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

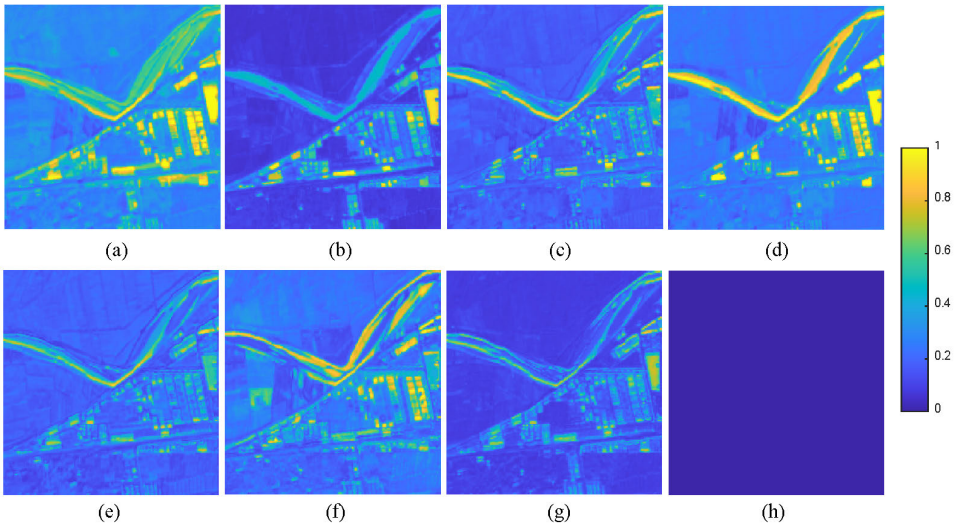


Fig. 15. SAM error images corresponding to Fig. 14 on the YRE dataset. (a) HS image at the auxiliary time of December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) GLMM-STSF. (f) USTSFM. (g) Proposed method. (h) HS image at target time of February 10, 2021 (reference).

is involved for discussion. As shown in Fig. 12, prominent changes are concentrated in the paddy fields and rivers on the right, and as a response, the spectrum of GLMM-STSF and USTSFM are not harmonious enough, while the spatial edge sharpening of Arad is still insufficient. It is noteworthy that LR-HS and HR fused images cannot directly generate SAM residual error plots, while the principle of keeping the original spectral structure requires a downsampling of the HR fused image to facilitate SAM calculation. Combined with SAM residual plots in Fig. 13, our method is capable of fully improving the image quality. In particular, DC-STSF significantly diminishes the spectral distortion of the background and also notices the aberration region in the lower right corner.

As for Fig. 14, the changed vegetation boundary in the lower right corner is noteworthy. Although Arad approaches reality in a spectrum, it is at the sacrifice of spatial details. In terms of subjective vision, Fig. 14(c)–(f) presents a fine performance in spatial quality, but leaves different degrees of spectral distortion in the anomalous region. Obviously, among all the methods, Arad acquires the best background spectral quality, while DC-STSF settles the error of some stubborn areas like river and paddy fields in Fig. 15.

Moreover, we utilize the reflectance index to measure the recovery precision of each independent band in Figs. 16 and 17. Also, of note, all fused images participating in the calculation pass the downsampling operator. We enumerate the reflectance curves on different types of landscapes as water, vegetation, and buildings. It can be found that time-varying land-cover changes reside in changes in curve profiles, and the traditional methods perform mediocrity under this aberration. The DL-based methods furnish better recovery quality, and our profile is closer to the reference value than USTSFM.

E. Ablation Studies

Ablation experiments were designed to prove the effectiveness of the proposed SAP module with designed components and the SFTN. In addition, the loss function terms were investigated individually to clarify the rationality of the design. The experiments were conducted on a reduced-resolution DY

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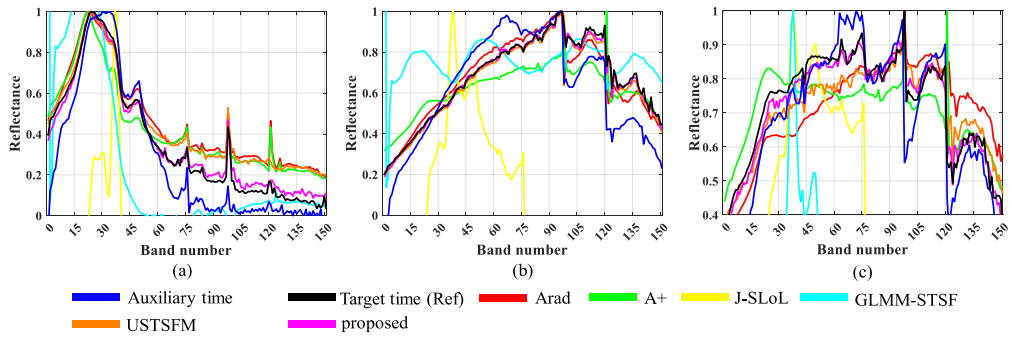


Fig. 16. Spectral curve comparison of typical surface features for different results corresponding to Fig. 12 on the DY dataset. (a) Water. (b) Vegetation. (c) Buildings.

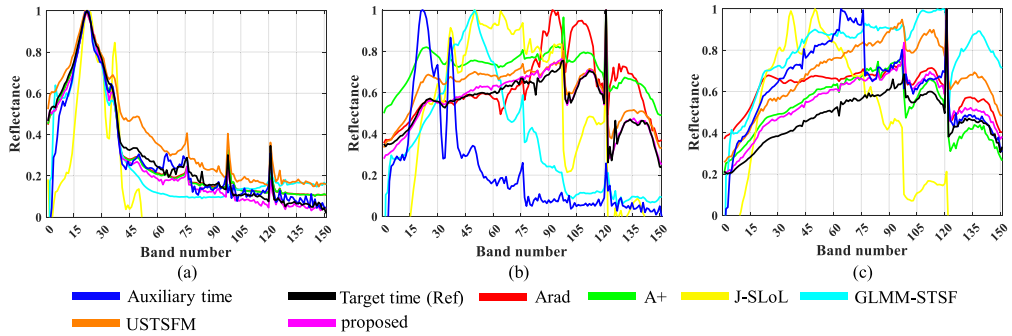


Fig. 17. Spectral curve comparison of typical surface features for different results corresponding to Fig. 14 on the YRE dataset. (a) Water. (b) Vegetation. (c) Buildings.

TABLE III

QUANTITATIVE EVALUATION RESULTS OF ABLATION STUDY ON THE DY REDUCED-RESOLUTION DATASETS. (THE BEST IS MARKED IN BOLD LETTERS AND THE SECOND-BEST IS UNDERLINED)

Model	PSNR	RMSE	ERGAS	SSIM	SAM	SCC
w/o SEU	<u>29.3724</u>	<u>0.0191</u>	3.1484	<u>0.9061</u>	<u>4.0340</u>	<u>0.8708</u>
w/o ECU	28.7937	0.0201	3.4042	0.9014	4.2972	0.8649
w/o SFTU	29.1084	0.0193	<u>3.0660</u>	0.8949	4.1682	0.8385
DC-STSF	**30.4601**	**0.0184**	**2.7436**	**0.9101**	**3.9915**	**0.8754**
N=2	29.2479	0.0201	3.1298	0.8734	4.3998	0.8636
N=3	**30.4601**	**0.0184**	**2.7436**	**0.9101**	**3.9915**	<u>0.8754</u>
N=4	<u>30.2395</u>	<u>0.0190</u>	<u>2.9382</u>	0.9006	4.1865	0.8695
N=5	29.3329	0.0194	2.9791	0.9048	4.1174	**0.8764**

TABLE IV

ABLATION STUDY FOR THE SELECTION OF LOSS FUNCTIONS ON THE DY REDUCED-RESOLUTION DATASETS. (THE BEST IS MARKED IN BOLD LETTERS AND THE SECOND-BEST IS UNDERLINED)

L_{MAE2}	L_{MAE1}	L_{grad}	L_{SAM}	PSNR	RMSE	ERGAS	SSIM	SAM	SCC
√	×	√	√	<u>29.6397</u>	<u>0.0188</u>	<u>2.9369</u>	<u>0.9071</u>	<u>4.1672</u>	<u>0.8736</u>
√	√	×	√	28.9532	0.0206	3.1467	0.8759	4.4160	0.8470
√	√	√	×	29.2759	0.0195	3.0139	0.8890	4.3794	0.8732
√	√	√	√	**30.4601**	**0.0184**	**2.7436**	**0.9101**	**3.9915**	**0.8754**

testing set, where the qualitative indicators are shown in Tables III and IV, and the corresponding visual results are shown in Fig. 18.

1) Effectiveness of SAP: SAP is considered to respect the real spatial–spectral details and effectively save computational cost. To determine the validity of SAP, we, respectively, removed the SEU and ECU and kept the overall structure unchanged. In Table III, the indices of the designed components are fully superior to the control items.

2) Number of Iterations of SAP: We certainly need to pick out the most appropriate iteration number from the perspective of parameter and performance. Table III illustrates that DC-STSF gets the optimal value when N is equal to 3.

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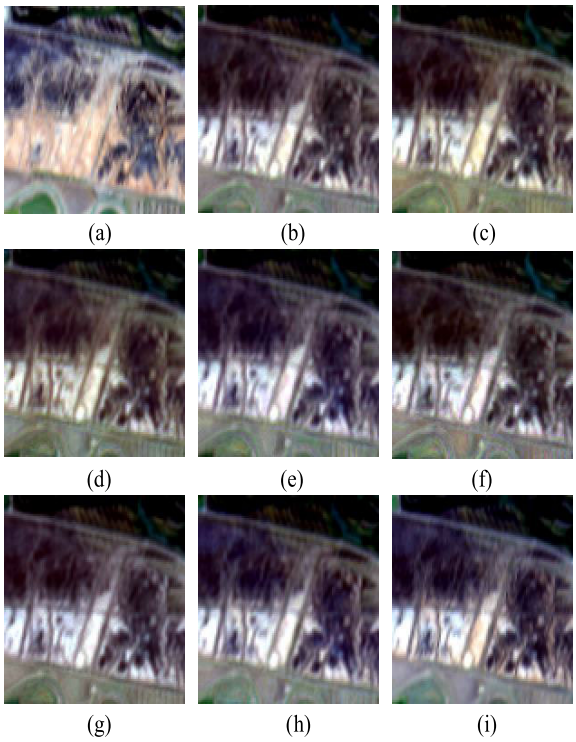


Fig. 18. Visual results of ablation studies on the DY dataset. (a) HS image at auxiliary time of December 11, 2019, (b) w/o ECU, (c) w/o SEU, (d) w/o SFTN, (e) w/o L_{MAE1}, (f) w/o L_{grad}, (g) w/o L_{SAM}, (h) DC-STSF, and (i) HS image at target time of February 10, 2021.

TABLE V
TESTING TIME OF REDUCED- AND FULL-RESOLUTION DATA ON THE DY DATASET

Method	Reduced-resolution	Full-resolution	Param
Arad	4.800s	6.526s	-
A+	3.614s	4.610s	-
J-SLoL	0.495s	1.209s	-
GLMM-STSF	46.992s	48.873s	-
USTSFM	0.0254s	0.0479s	7490k
DC-STSF	**0.0189s**	**0.0418s**	**5721k**

3) Effectiveness of the SFTN: The location of the SFTN is to derive rough spectral characteristics of target time through obtained spectral knowledge. To clarify whether the SFTN really performs our expected function, we replaced this part with the convolution of concatenated F_1 and MS_2. Table III shows that the indices dropped sharply in the PSNR and SCC in the control term, which explains that the adaptive fine-tuning is conducive to the recognition of complex land cover.

4) Selection of Loss Functions: The loss functions were determined from the perspective of spatial, spectral, and global quality. Allowing for the necessity of L_{MAE2}, we tested the remaining constraints, in turn, and the corresponding results are shown in Table IV. The indicators in the first column prove that the lack of L_{MAE1} leads to a significant decline in the overall performance. The basis is crucial because it directly proves the authentic interdependence between SSJE and TVE tasks, which follows the design concept of our network. The indicators in the second column prove that the absence of L_{grad} makes a sharp decrease in SCC, which also reflects that L_{grad} has provided help of spatial sharpening. Similarly, the third line proves the value of L_{SAM} in spectral fidelity through the SAM indicator. The above experiments show that the selection of constraint terms in the training is reasonable.

F. Parameters and Testing Time

In this section, the complexity and operation efficiency of these comparison methods were investigated. The tensors with $1 \times 4 \times 72 \times 72$ and $1 \times 151 \times 24 \times 24$ were applied to the HR-MS and LR-HS roles for evaluation, respectively. The results on reduced- and full-resolution data are shown in Table V. It shows that our proposed method achieves competitive performance.

IV. CONCLUSION

In this article, we proposed a dual-task cascaded spatial–temporal–spectral image fusion method with SSJE and TVE modeling into a unified framework. The SSJE with an iterative alternating projection module was designed. The SEU and ECU on feedback fusion of errors were further developed to gain potential. The TVE on the SFTN was developed, by considering the inconsistent spectral overlap range between HR-MS and LR-HS images, to further enhance the temporal and spectral fidelities. The experiments conducted on the reduced- and full-resolution DY and YRE datasets verify that our method achieves superior performance over the existing methods qualitatively and quantitatively. In addition, considering that the noise in the LR-HS images brings uncertainty to the rebuild of the spatial–spectral features, the denoising and fusion joint model in the spatial-frequency domain can be researched.

REFERENCES

[1] C. Van Westen, "Remote sensing for natural disaster management," *Int. Arch. Photogramm. Remote Sens.*, vol. 33, pp. 1609–1617, Jul. 2000.
[2] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.
[3] G. M. Foody, "Status of land cover classification accuracy assessment," *Remote Sens. Environ.*, vol. 80, no. 1, pp. 185–201, Apr. 2002.
[4] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.
[5] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.
[6] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.
[7] G. Yang, M. Zhou, K. Yan, A. Liu, X. Fu, and F. Wang, "Memory-augmented deep conditional unfolding network for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1778–1787.

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[8] D. Wang, Y. Bai, C. Wu, Y. Li, C. Shang, and Q. Shen, "Convolutional LSTM-based hierarchical feature fusion for multispectral pan-sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404016.

[9] M. Zhou, K. Yan, J. Huang, Z. Yang, X. Fu, and F. Zhao, "Mutual information-driven pan-sharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1798–1808.

[10] Q. Liu, H. Zhou, Q. Xu, X. Liu, and Y. Wang, "PSGAN: A generative adversarial network for remote sensing image pan-sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10227–10242, Dec. 2021.

[11] H. Lu, Y. Yang, S. Huang, W. Tu, and W. Wan, "A unified pansharpening model based on band-adaptive gradient and detail correction," *IEEE Trans. Image Process.*, vol. 31, pp. 918–933, 2022.

[12] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 3, no. 3, pp. 27–46, Sep. 2015, doi: 10.1109/MGRS.2015.2440094.

[13] X. Meng, H. Shen, H. Li, L. Zhang, and R. Fu, "Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges," *Information Fusion*, vol. 46, pp. 102–113, Mar. 2019.

[14] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008, doi: 10.1109/TGRS.2008.916211.

[15] B. Aiazzi, L. Alparone, S. Baronti, and M. Selva, "MS + pan image fusion by enhanced gram-Schmidt spectral sharpening," in *in Proc. 26th EARSel Symp.*, May 2006, pp. 1–12.

[16] X. Otazu, M. González-Audícana, O. Fors, and J. Nú nez, "Introduction of sensor spectral response into image fusion methods. Application to wavelet-based methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 10, pp. 2376–2385, Oct. 2005.

[17] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and pan imagery," *Photogrammetric Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[18] G. Vivone et al., "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[19] F. Fang, F. Li, C. Shen, and G. Zhang, "A variational approach for pan-sharpening," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2822–2834, Jul. 2013.

[20] F. Ozcelik, U. Alganci, E. Sertel, and G. Unal, "Rethinking CNN-based pansharpening: Guided colorization of panchromatic images via GANs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3486–3501, Apr. 2021.

[21] D. Lei, H. Chen, L. Zhang, and W. Li, "NLRNet: An efficient nonlocal attention ResNet for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5401113.

[22] L.-J. Deng, G. Vivone, C. Jin, and J. Chanussot, "Detail injection-based deep convolutional neural networks for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6995–7010, Aug. 2021, doi: 10.1109/TGRS.2020.3031366.

[23] Z. Wang, B. Chen, R. Lu, H. Zhang, H. Liu, and P. K. Varshney, "FusionNet: An unsupervised convolutional variational network for hyperspectral and multispectral image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 7565–7577, 2020.

[24] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, 2016.

[25] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, and J. Paisley, "PanNet: A deep network architecture for pan-sharpening," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 5449–5457.

[26] X. Liu, Q. Liu, and Y. Wang, "Remote sensing image fusion based on two-stream fusion network," *Inf. Fusion*, vol. 55, pp. 1–15, Mar. 2020.

[27] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[28] M. Sim oes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[29] W. Sun et al., "MLR-DBPFN: A multi-scale low rank deep back projection fusion network for anti-noise hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[30] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Reciprocal transformer for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102148.

[31] W. Li, X. Zhang, Y. Peng, and M. Dong, "Spatiotemporal fusion of remote sensing images using a convolutional neural network with attention and multiscale mechanisms," *Int. J. Remote Sens.*, vol. 42, no. 6, pp. 1973–1993, Mar. 2021.

[32] W. Li, X. Zhang, Y. Peng, and M. Dong, "DMNet: A network architecture using dilated convolution and multiscale mechanisms for spatiotemporal fusion of remote sensing images," *IEEE Sensors J.*, vol. 20, no. 20, pp. 12190–12202, Oct. 2020.

[33] Z. Tan, M. Gao, X. Li, and L. Jiang, "A flexible reference-insensitive spatiotemporal fusion model for remote sensing images using conditional generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[34] H. Song, Q. Liu, G. Wang, R. Hang, and B. Huang, "Spatiotemporal satellite image fusion using deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 821–829, Mar. 2018.

[35] D. Jia, C. Cheng, C. Song, S. Shen, L. Ning, and T. Zhang, "A hybrid deep learning-based spatiotemporal fusion method for combining satellite images with different resolutions," *Remote Sens.*, vol. 13, no. 4, p. 645, 2021.

[36] I. V. Emelyanova, T. R. McVicar, T. G. Van Niel, L. T. Li, and A. I. J. M. van Dijk, "Assessing the accuracy of blending Landsat–MODIS surface reflectances in two landscapes with contrasting spatial and temporal dynamics: A framework for algorithm selection," *Remote Sens. Environ.*, vol. 133, pp. 193–209, Jun. 2013.

[37] X. Zhu, F. Cai, J. Tian, and T. Williams, "Spatiotemporal fusion of multisource remote sensing data: Literature survey, taxonomy, principles, applications, and future directions," *Remote Sens.*, vol. 10, no. 4, p. 527, Mar. 2018.

[38] Z. Niu, "Use of MODIS and Landsat time series data to generate high-resolution temporal synthetic Landsat data using a spatial and temporal reflectance fusion model," *J. Appl. Remote Sens.*, vol. 6, no. 1, Mar. 2012, Art. no. 063507.

[39] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006.

[40] X. Zhu, J. Chen, F. Gao, X. Chen, and J. G. Masek, "An enhanced spatial and temporal adaptive reflectance fusion model for complex heterogeneous regions," *Remote Sens. Environ.*, vol. 114, no. 11, pp. 2610–2623, Nov. 2010.

[41] H. Zhu et al., "Fusion of multisource satellite AOD products via Bayesian maximum entropy with explicit a priori knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[42] B. Huang and H. Song, "Spatiotemporal reflectance fusion via sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3707–3716, Oct. 2012.

[43] X. Liu, C. Deng, J. Chanussot, D. Hong, and B. Zhao, "StfNet: A two-stream convolutional neural network for spatiotemporal image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6552–6564, Sep. 2019.

[44] W. Li, D. Cao, Y. Peng, and C. Yang, "MSNet: A multi-stream fusion network for remote sensing spatiotemporal fusion based on transformer and convolution," *Remote Sens.*, vol. 13, no. 18, p. 3724, Sep. 2021.

[45] B. Chen, B. Huang, and B. Xu, "Constucting a unified framework for multi-source remotely sensed data fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 2574–2577.

[46] Y. Zhao and B. Huang, "Integrating modis and MTSAT-2 to generate high spatial–temporal-spectral resolution imagery for real-time air quality monitoring," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 6122–6125.

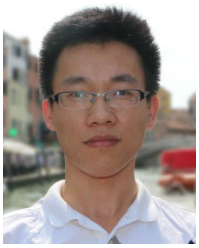
[47] H. Shen, "Integrated fusion method for multiple temporal–spatial–spectral images," in *Proc. 22nd Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, Melbourne, VIC, Australia, Aug. 2012, pp. 407–410.

[48] B. Huang, H. Zhang, H. Song, J. Wang, and C. Song, "Unified fusion of remote-sensing imagery: Generating simultaneously high-resolution synthetic spatial–temporal–spectral Earth observations," *Remote Sens. Lett.*, vol. 4, no. 6, pp. 561–569, Jun. 2013.

[49] H. Shen, X. Meng, and L. Zhang, "An integrated framework for the spatio–temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7135–7148, Dec. 2016.

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[50] Y. Peng, W. Li, X. Luo, J. Du, Y. Gan, and X. Gao, "Integrated fusion framework based on semicoupled sparse tensor factorization for spatio-temporal–spectral fusion of remote sensing images," *Inf. Fusion*, vol. 65, pp. 21–36, Jan. 2021.
[51] J. Zhou, W. Sun, X. Meng, G. Yang, K. Ren, and J. Peng, "Generalized linear spectral mixing model for spatial–temporal–spectral fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[52] X. Chen, X. Meng, F. Shao, and W. Sun, "PSSTFN: A progressive spatial–temporal–spectral fusion network for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5530312.
[53] Q. Liu, X. Chen, X. Meng, H. Chen, F. Shao, and W. Sun, "Dual-task interactive learning for unsupervised spatio-temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402015.
[54] M. Haris, G. Shakhnarovich, and N. Ukita, "Deep back-projection networks for super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1664–1673.
[55] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 19–34.
[56] J. Wu, J. Aeschbacher, and R. Timofte, "In defense of shallow learned spectral reconstruction from RGB images," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 471–479.
[57] L. Gao, D. Hong, J. Yao, B. Zhang, P. Gamba, and J. Chanussot, "Spectral superresolution of multispectral imagery with joint sparse and low-rank learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2269–2280, Mar. 2021.
[58] Z. Wang and A. C. Bovik, "Mean squared error: Love it or leave it? A new look at signal fidelity measures," *IEEE Signal Process. Mag.*, vol. 26, no. 1, pp. 98–117, Jan. 2009.
[59] T. O. Hodson, "Root mean square error (RMSE) or mean absolute error (MAE): When to use them or not," *Geosci. Model Dev. Discuss.*, vol. 15, no. 14, pp. 1–10, Mar. 2022.
[60] L. Wald, *Data Fusion: Definitions and Architectures: Fusion of Images of Different Spatial Resolutions*. France, Paris: Presses des MINES, 2002.
[61] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[62] R. H. Yuhas, A. F. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. 3rd Annual JPL Airborne Geosci. Shop*, vol. 1, Jun. 1992.
[63] X. Chen et al., "FTDN: Multispectral and hyperspectral image fusion with diverse temporal difference spans," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, pp. 5530312:1–5530312:12.
[64] Q. Liu, X. Meng, S. Zhang, X. Li, and F. Shao, "A temporally insensitive spatio-temporal fusion method for remote sensing imagery via semantic prior regularization," *Inf. Fusion*, vol. 117, May 2025, Art. no. 102818.



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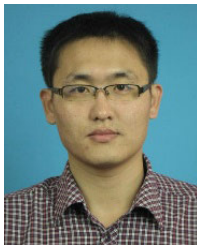
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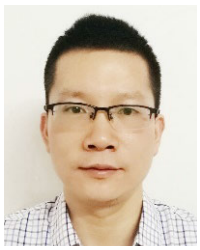
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